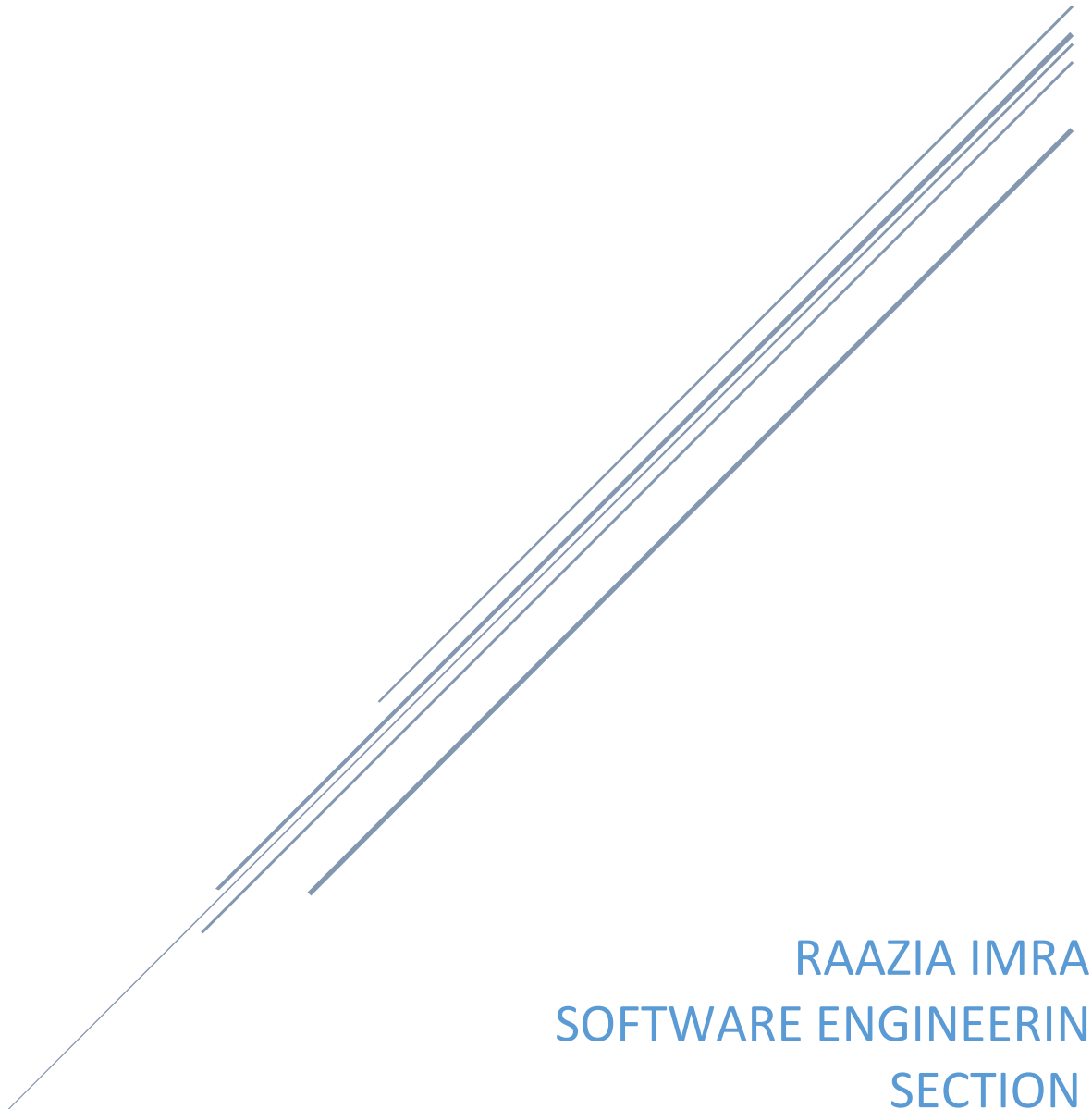


ASSIGNMENT#1

PROGRAMMING FUNDAMENTALS

CT-175

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SOFTWARE ENGINEERING
SECTION A
SE-O9

Assignment No:1
Programming Fundamentals
FESE- Software Engineering Department

CLO-2(C3)

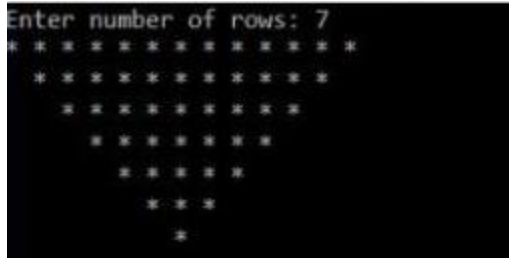
Due Date:27th Sep,24

Apply the devised solutions into computer programs and test the programs on a computer.

It should be clear that your assignment will not get any credit if:

- The assignment is submitted after due date.
- The assignment is submitted via email.
- The assignment is copied from Internet or from any other student.

Q.1 Write a python program following any Sequence structure to generate this output.



```
Enter number of rows: 7
* * * * *
 * * * * *
  * * * * *
   * * * * *
    * * * * *
     * * * * *
      * * * * *
```

PROGRAM

```
1  # Ask the user to enter the number of rows for the pyramid
2  n = int(input("Enter number of rows: "))
3
4  # Loop to print each row of the pattern
5  for i in range(n, 0, -1):
6      spaces = ' ' * (n-i) # Calculate spaces to center the stars
7      stars = '* ' * (2 * i - 1) # Adjust the number of stars
8      print(spaces + stars)
9
```

OUTPUT

```
C:\Users\hp\PycharmProjects\
Enter number of rows: 7
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

Q2 By using For loop in python program, generate the following series.

a. 11 16 21 26 31 36 41

PROGRAM

```
1 # Start at 11 and increase by 5 each time
2 for i in range(11, 42, 5):
3     print(i, end=' ')
```

OUTPUT

```
C:\Users\hp\PycharmProjects\main\ven'
11 16 21 26 31 36 41
Process finished with exit code 0
```

b. 3 8 13 18 23 28 33

PROGRAM

```
1  # Start at 3 and increase by 5 each time
2  for i in range(3, 34, 5):
3      print(i, end=' ')
```

OUTPUT

```
↑ C:\Users\hp\PycharmProjects\main\ven
↓ 3 8 13 18 23 28 33
⇨ Process finished with exit code 0
```

C. 20 20 17 17 14 14 11

PROGRAM

```
1  # Initialize the starting number
2  start = 20
3
4  # Use a loop to generate the sequence
5  for i in range(7): # We need 7 elements in total
6      print(start, end=' ')
7      # After every two iterations, decrease the number by 3
8      if i % 2 == 1:
9          start -= 3
```

OUTPUT

```
C:\Users\hp\PycharmProjects\main\v
20 20 17 17 14 14 11
Process finished with exit code 0
```

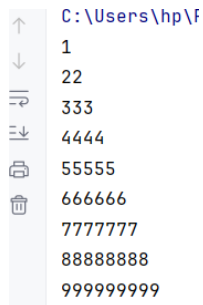
Q.3 Write a Python program to construct the following pattern, using a nested for loop

```
1
22
333
4444
55555
666666
7777777
88888888
999999999
```

PROGRAM

```
1  # Number of rows for the pattern
2  rows = 9
3
4  # Outer loop for each row
5  for i in range(1, rows + 1):
6      # Inner loop for printing the current row number i, i times
7      for j in range(i):
8          print(i, end='') # Print the current row number without a space
9      print() # Move to the next line after each row
```

OUTPUT



C:\Users\hp\F
1
22
333
4444
55555
666666
7777777
88888888
999999999

Q.5 Write a Python program that prompts for amount in rupees and show how many 1000's, 500's, 100's, 50's, 10's, 5's, 2's and 1's in it.

Sample run:

Enter amount in rupees: 5788
1000's in the given amount is: 5
500's in the given amount is: 1
100's in the given amount is: 2
50's in the given amount is: 1
10's in the given amount is: 3
5's in the given amount is: 1
2's in the given amount is: 2
1's in the given amount is: 1

PROGRAM

```
1  def count_notes(amount): 1 usage
2      denominations=(1000,500,100,50,10,5,2,1)
3      notes={}
4      for denomination in denominations:
5          notes[denomination]=amount//denomination
6          amount%=denomination
7      return notes
8  amount=int(input("Enter the amount in rupees:"))
9  note_counts=count_notes(amount)
10 for denomination,count in note_counts.items():
11     print(f"{denomination}'s in the given amount is:{count}")
```

OUTPUT

```
C:\Users\hp\PycharmProjects\main\venv\S
Enter the amount in rupees:6798
1000's in the given amount is:6
500's in the given amount is:1
100's in the given amount is:2
50's in the given amount is:1
10's in the given amount is:4
5's in the given amount is:1
2's in the given amount is:1
1's in the given amount is:1

Process finished with exit code 0
```

Q.6 Write a python program to perform the following conditional actions:

Given an integer,

- If is odd, print Karachi
- If is even and in the inclusive range of 2 to 5 , print Lahore
- If is even and in the inclusive range of 6 to 20 , print Islamabad
- If is even and greater than 20 , print Pakistan

PROGRAM

```
1  # Prompt the user to enter an integer
2  number = int(input("Enter an integer: "))
3
4  # Conditional actions based on the value of the integer
5  if number % 2 == 1: # Check if the number is odd
6      print("Karachi")
7  elif number % 2 == 0: # Check if the number is even
8      if 2 <= number <= 5: # Check if the number is in the range 2 to 5
9          print("Lahore")
10     elif 6 <= number <= 20: # Check if the number is in the range 6 to 20
11         print("Islamabad")
12     elif number > 20: # Check if the number is greater than 20
13         print("Pakistan")
```

OUTPUT

```
C:\Users\hp\PycharmProj
Enter an integer: 3
Karachi
```

```
C:\Users\hp\PycharmPro
Enter an integer: 24
Pakistan
```



```
C:\Users\hp\PycharmProj
Enter an integer: 2
Lahore
```

```
C:\Users\hp\PycharmPr
Enter an integer: 6
Islamabad
```

Q.7 Write a python program that takes age as an input and check If age is greater than or equal to 18, it checks the second condition for checking if one has license or not, If has_license is True, the person can drive, otherwise he can't.

PROGRAM

```
1  # Prompt the user for their age
2  age = int(input("Enter your age: "))
3
4  # Check if the age is greater than or equal to 18
5  if age >= 18:
6      has_license = input("Do you have a driver's license? (yes/no): ").strip().lower()
7
8      # Check the license status
9      if has_license == "yes":
10         print("You can drive.")
11     else:
12         print("You can't drive.")
13 else:
14     print("You can't drive because you are under 18.")
```

OUTPUT

```
C:\Users\hp\PycharmProjects\main\venv\Scripts'
Enter your age: 19
Do you have a driver's license? (yes/no): yes
You can drive.
```

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe
```

```
Enter your age: 16
```

```
You can't drive because you are under 18.
```

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe
```

```
Enter your age: 23
```

```
Do you have a driver's license? (yes/no): no
```

```
You can't drive.
```

Q.8 Given a list of numbers holding numbers = [5, 10, -3, 7, -2, 9, -8, 4, -11]

- Print some message showing Skips processing negative numbers.
- terminates the loop when a number greater than 10 is encountered.
- Print ,All non-negative numbers less than or equal to 10 are processed.

PROGRAM

```
1 numbers = [5, 10, -3, 7, -2, 9, -8, 4, -11]
2 # Process each number in the list
3 for num in numbers:
4     if num < 0:
5         print(f"Skipping processing negative number: {num}")
6         continue # Skip to the next iteration for negative numbers
7
8     if num > 10:
9         print("Terminating loop since a number greater than 10 is encountered.")
10        break # Exit the loop if a number greater than 10 is found
11
12    # Print processed numbers that are non-negative and less than or equal to 10
13    print(f"Processing number: {num}")
14
15 # Final message
16 print("All non-negative numbers less than or equal to 10 are processed.")
```

OUTPUT

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe C:\Users\hp\PycharmProjects\main\MAIN.py
Processing number: 5
Processing number: 10
Skipping processing negative number: -3
Processing number: 7
Skipping processing negative number: -2
Processing number: 9
Skipping processing negative number: -8
Processing number: 4
Skipping processing negative number: -11
All non-negative numbers less than or equal to 10 are processed.
```

Q.9 Write a python program consisting of following conditions

- The loop allows the user to withdraw money until the balance becomes 0 or the number of attempts reaches the withdrawal limit.
- If the user tries to withdraw more than the available balance, it prints a message and skips the iteration (`continue`).
- If the balance reaches zero, the loop ends(`break`)

PROGRAM

```
1  # Initial balance and withdrawal limit
2  balance = 1000 # Starting balance
3  withdrawal_limit = 3 # Maximum number of withdrawal attempts
4  attempts = 0 # Counter for the number of withdrawal attempts
5
6  # Loop to allow withdrawals
7  while attempts < withdrawal_limit:
8      # Prompt user for withdrawal amount
9      withdrawal_amount = float(input("Enter amount to withdraw: "))
10
11     # Check if the withdrawal amount is greater than the available balance
12     if withdrawal_amount > balance:
13         print(f"Insufficient balance! You can only withdraw up to {balance}.")
14         continue # Skip to the next iteration if insufficient funds
15
16     # Deduct the withdrawal amount from the balance
17     balance -= withdrawal_amount
18     attempts += 1 # Increment the attempt counter
19
20     # Print the current balance
21     print(f"Withdrawal successful! Your new balance is: {balance}")
22
23     # Check if the balance has reached zero
24     if balance == 0:
25         print("Your balance is now zero. Exiting.")
26         break # Exit the loop if the balance is zero
27
28     # Final message if withdrawal attempts are exhausted
29     if attempts >= withdrawal_limit:
30         print("Withdrawal limit reached. You cannot withdraw any more money.")
```

OUTPUT

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe C:\Users\hp\PycharmProjects\main\MAIN.py
```

```
Enter amount to withdraw: 560
```

```
Withdrawal successful! Your new balance is: 440.0
```

```
Enter amount to withdraw: 345
```

```
Withdrawal successful! Your new balance is: 95.0
```

```
Enter amount to withdraw: 100
```

```
Insufficient balance! You can only withdraw up to 95.0.
```

```
Enter amount to withdraw: 95
```

```
Withdrawal successful! Your new balance is: 0.0
```

```
Your balance is now zero. Exiting.
```

```
Withdrawal limit reached. You cannot withdraw any more money.
```

```
Process finished with exit code 0
```